UNIT-4: Express JS

Introduction

Express.js is a small framework that works on top of Node.js web server functionality to simplify its APIs and add helpful new features. It makes it easier to organize your application's functionality with middleware and routing. It adds helpful utilities to Node.js HTTP objects and facilitates the rendering of dynamic HTTP objects.

Features

- Develops Node.js web applications quickly and easily.
- It's simple to set up and personalize.
- It can be used to design single-page, multi-page and hybrid web applications.
- Allows you to define application routes using HTTP methods and URLs.
- Includes a number of middleware modules that can be used to execute additional requests and responses activities.
- Simple to interface with a variety of template engines, including Jade, Vash, and EJS.
- Allows you to specify a middleware for handling errors.

Using Express.js, one can create REST API 5 times faster & using less Code.

Environment Setup

Node Package Manager(npm)

npm is the package manager for node. The npm Registry is a public collection of packages of open-source code for Node.js, front-end web apps, mobile apps, robots, routers, and countless other needs of the JavaScript community. npm allows us to access all these packages and install them locally. You can browse through the list of packages available on npm at npmJS.

How to use npm?

There are two ways to install a package using npm: globally and locally.

Globally − This method is generally used to install development tools and CLI based packages. To install a package globally, use the following code.

npm install -g <package-name>

Locally − This method is generally used to install frameworks and libraries. A locally installed package can be used only within the directory it is installed. To install a package locally, use the same command as above without the **-g** flag.

npm install <package-name>

Whenever we create a project using npm, we need to provide a **package.json** file, which has all the details about our project. npm makes it easy for us to set up this file. Let us set up our development project.

Step 1 − Start your terminal/cmd, create a new folder named Express and cd (create directory) into it

- mkdir Express
- cd Express
- code Filename

Step 2 − Now to create the package.json file using npm, use the following code.

npm init

It will ask you for the following information.

```
{ About to write to D:\FSD-2\psp\package.json:
{
  "name": "express",
  "version": "1.0.0",
  "description": "First express",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "author": "Priyen Patel",
  "license": "ISC"
}
```

Note: By using **npm init -y** it will set automatic values in above fields.

Step 3 − Now we have our package.json file set up, we will further install Express. To install Express and add it to our package.json file, use the following command −

npm install express –save OR npm install express **OR** npm i express

The **--save** flag can be replaced by the **-S** flag. This flag ensures that Express is added as a dependency to our **package.json** file. This has an advantage, the next time we need to install all the dependencies of our project we can just run the command *npm install* and it will find the dependencies in this file and install them for us.

This will generate package-lock.json file.

This is all we need to start development using the Express framework.

Command to check version of express: **npm list express**

Express Methods

For routing with different pages, it is too easy than Node.js. There is no need to write any if-else ladder to compare requested page.

Step-1 Import Express and create application

```
const expr=require("express")

const app=expr();
```

- Importing the Express module.
- express() is a function that creates an instance of an Express application.
- Creating an app instance (like creating a server) that you can use to define routes, middleware, etc.

Step-2

Routing refers to determining how an application responds to a client request to a particular endpoint, which is a URI (or path) and a specific HTTP request method (GET, POST, etc.).

Note: The main difference between a URL and a URI is that a URL specifies the location of a resource on the internet, while a URI can be used to identify any type of resource, not just those on the internet.

Each route can have one or more handler functions, which are executed when the route is matched. Route definition takes the following structure:

app.METHOD(PATH, HANDLER)

Where:

- app is an instance of express.
- METHOD is an HTTP request method, in lowercase.
- PATH is a path on the server.
- HANDLER is the function executed when the route is matched.

The following examples illustrate defining simple routes.

Method -1 : app.get(route, callback)

This function tells what to do when a **get** request at the given route is called. The callback function has 2 parameters, ***request(req)*** and ***response(res)***.
The request **object(req)** represents the HTTP request and has properties for the request query string, parameters, body, HTTP headers, etc. Similarly, the response object represents the HTTP response that the Express app sends when it receives an HTTP request.

Respond with Hello World! on the homepage:

```
app.get('/', (req, res) => {
  res.send('Hello World!')
})
```

(**callback**- An asynchronous function that is called when the server starts listening for requests.).

Method 2: app.post(route, callback)

Respond to POST request on the root route (/), the application's home page:

```
app.post('/', (req, res) => {
  res.send('Got a POST request')
})
```

Note:
The **app.all()** function is used to route all types of HTTP requests. Like if we have POST, GET, PUT, DELETE, etc, requests made to any specific route, let's say */user*, so instead of defining different APIs like app.post('/user'), app.get('/user'), etc, we can define single API **app.all('/user')** which will accept all type of HTTP request.

Step-3 Response methods

The methods on the response object (res) in the following table can send a response to the client, and terminate the request-response cycle.

Method	Code Example	Use Case
res.set()	**res.set("content-type", "text/plain")**	**Manually set headers**
res.type()	res.type("text/plain")	Set content type
res.setHeader()	res.setHeader("Content-type", "text/plain")	Node.js style
res.write("string")	res.write("Hello")	function to write something on browser.
res.send()	res.send()	Set header automatically and close connection. *Must be blank if used with res.write()

- **res.set()**
  This function is used to set the response HTTP header field to value. To set multiple fields at once, pass an object as the parameter.

  Syntax: res.set(field , value)

  Parameters: The field parameter is the name of the field and the value parameter is the value assigned to the field parameter.
  Return Value: It returns an Object.

- **res.send()**
  - Since send( ) function can be called once and its like response is sent to client, now nothing can be happened. So after res.send() method, other send() or write() method can not be used. send() method is equivalent to end() method of Node.JS.
  - If you pass in a string, it sets the Content-Type header to text/html .
  - if you pass in an object or an array, it sets the application/json Content-Type header, and parses that parameter into JSON.
  - send() automatically sets the Content-Length HTTP response header.
  - send() also automatically closes the connection.

Response methods

The methods on the response object (res) in the following table can send a response to the client, and terminate the request-response cycle

Method	Description	Syntax	Used for
res.send()	Send a response of various types.	res.send("Hello")	Closing connection + Automatic set header
res.json()	Send a JSON response and close connection*.	res.json({ name: "A" })	APIs
res.sendFile()	Send a file as an octet stream.	res.sendFile("a.html")	File downloads or serving
res.redirect()	Redirect a request to another URL	res.redirect("/payment")	Navigation or routing
res.render()	Render a view template.	res.render("index")	Server-side rendering
res.end()	End the response process.	res.end()	Node concept. Minimal response handling

Basic Note:

- **res.end:** comes from NodeJS core. In Express JS if you need to end request in a quick way and do not need to send any data then you can use this function.
- **res.send:** Sends data and end the request.
- **res.json:** Sends data in JSON format and ends the request.
- **res.json():** allows for extra formatting of the JSON data - if this is not required **res.send()** can also be used to return a response object using Express. Both of these methods also end the response correctly, and there's no further action required.

Step-4

app.listen(port, [host], [backlog], [callback]])

This function binds and listens for connections on the specified host and port. Port is the only required parameter here.

Sr No.	Argument & Description
1	**port** A port number on which the server should accept incoming requests.
2	**host** Name of the domain. You need to set it when you deploy your apps to the cloud.
3	**backlog** The maximum number of queued pending connections. The default is 511.
4	**callback** An asynchronous function that is called when the server starts listening for requests.

Example: app.listen(3000, 'localhost', 100, () => {console.log('Server started');});

Example:

Index.js

```
var express = require('express');
var app = express();
app.get('/', function(req, res){
 res.set ("content-type","text/plain");
 res.send("Hello world!");

});
app.listen(3000);
```

Save the file, go to your terminal and type the following.

node index.js

This will start the server. To test this app, open your browser and go to **http://localhost:3000** and a message will be displayed **Hello world!**

Routing

Introduction to Routing

- ✓ In Express.js, routing refers to the process of determining how an application responds to a client request for a specific endpoint.
- ✓ An endpoint is defined by a URI (Uniform Resource Identifier) or path along with an HTTP request method such as GET, POST, PUT, or DELETE.
- ✓ Routing is one of the core features of Express.js and is used to build APIs and web applications.
- ✓ Each route in Express can have one or more handler functions. These handler functions are executed when a request matches the defined route.

Difference between URL and URI
A URL (Uniform Resource Locator) is used to specify the location of a resource on the internet. It provides information about where the resource is located and how to access it.
A URI (Uniform Resource Identifier) is a more general concept that is used to identify any resource, not only those available on the internet. A URL is a type of URI.

Example:
API in Express.js
Consider an API endpoint in Express.js
GET /users/101
Here, /users/101 is a **URI** because it identifies a specific user resource.
If this endpoint is accessed using
http://localhost:3000/users/101
then it becomes a **URL** because it includes the protocol and location needed to access the resource.

Structure of Route Definition

Route definition takes the following structure:

app.METHOD(PATH, HANDLER)

- **app** is an instance of express.
- **METHOD** is an HTTP request method, in lowercase.
- **PATH** is a route path on the server.
- **HANDLER** is the callback function that executes when the route is matched.

Routing using HTTP Methods

HTTP methods define the type of operation that should be performed on a resource. The most commonly used HTTP methods in Express.js are GET, POST, PUT, and DELETE.

- ✓ GET is used to retrieve data from the server.
- ✓ POST is used to send data to the server.
- ✓ PUT is used to update existing data.
- ✓ DELETE is used to remove data from the server.

Example of a GET request on the homepage:

```
app.get('/', (req, res) => {
res.send('Hello World!')
})
```

In this example, when a client sends a GET request to the root URL, the server responds with "Hello World".

Example of a POST request on the homepage:

```
app.post('/', (req, res) => {
res.send('Got a POST request')
})
```

In this case, when a POST request is made to the same path, a different response is sent. These examples show that the same route can behave differently depending on the HTTP method used.

Handling Multiple HTTP Methods using app.all()

The app.all() function in Express.js is used to handle all types of HTTP requests for a specific route. This means it will respond to GET, POST, PUT, DELETE, and any other HTTP method.

```
Example:
app.all('/user', (req, res) => {
res.send('This route handles all HTTP methods')
})
```

Using app.all() helps reduce code duplication when the same logic needs to be applied to multiple request methods.

Key Points about Routing and HTTP Methods

- ✓ Routing connects client requests to server responses.
- ✓ Each route is defined using a method, path, and handler function.
- ✓ Different HTTP methods can be used on the same route to perform different operations.
- ✓ The app.all() method allows handling all HTTP methods with a single route definition.
- ✓ Routing is essential for building RESTful APIs and web applications.

*** In Node JS we already learnt http module to route on different pages.**
(Ref of Node only*)

```
var http=require("http");
var server=http.createServer( (req,res) =>{
   if(req.url=="/")
   {  res.writeHead(200,{"content-type":"application/json"});
      res.write("<h1>Thank you..!</h1>");
      res.end();    }
else if(req.url=="/student"){
   res.writeHead(200,{"content-type":"text/plain"}); //plain shows code as it is
   res.write("<i> Home page1 </i>");
   res.end();}});
server.listen(6001);
```

How the Express Routing Works?

Example

p1.js	**Output**

const expr = require("express");
const app = expr();
app.get ("/", (req,res)=>{
res.set ("content-type","text/plain");
res.send ("<h1>Hello</h1>");
});
app.get ("/about", (req,res)=>{
res.set ("content-type","text/html");
res.write ("Hello");
res.send ();
//res.write ("hello");
});
app.listen (5504,()=>
{ console.log ("server started");
})
	In terminal > node p1.js
	Console >>
	server started
	In browser >>
	• localhost:5504
	<h1>Hello</h1>
	• localhost:5504/about
	Hello
	Note: Like node JS we cannot set headers after they are sent to the client. Means if we send response by writing in "res.write()" then "res.send()" must be blank.
	
	// Res.write() after the res.send() will generate an error in terminal "write after end".(Commented line)

Json Response Passing

We can define a JSON object and directly pass it inside res.send(). For this, JSON.stringify() method is not required. To send JSON object in res.write(), convert JSON object to string using JSON.stringify().

Example : Write Express JS script to request server to display json object on browser.

Send JSON object using res.write	**Send JSON object using res.send**

const expr=require("express")
const app=expr();
student={name:"LJU",age:28}
app.get("/",(req,res)=>{
res.set("content-type", "application/json")
res.write(JSON.stringify(student))
res.send()
})
app.listen(6007)
	const expr=require("express")
	const app=expr();
	student={name:"LJU",age:28}
	app.get("/",(req,res)=>{
	res.set("content-type","application/json")
	res.send(student)//res.send automatically converts data in json format
	})
	app.listen(6007)

Output: {"name":"LJU","age":28}	**Output:** {"name":"LJU","age":28}

You can send JSON to the client by using **res.json()** , a useful method.

It accepts an object or array, and converts it to JSON before sending it:

res.json(student) in above case also work. It will send data and close the connection.

Write express script to send json object to server and display it.

Send JSON object using res.json

```
const express=require("express")
const app=express();
student={name:"LJU",age:28}
  app.get("/",(req,res)=>{
    res.json(student)
  })
app.listen(6007)
```

Output in browser: {"name":"LJU","age":28}

Example : Write Express JS script to request server to display only Age on browser.

```
const express=require("express")
const app=express();
student={name:"LJU",age:28}
   app.get("/",(req,res)=>{
      res.write(student.age+"")   //needs to convert to string so appended string
      res.send()
   })
   app.get("/j",(req,res)=>{
      res.send(student.age)  //needs to convert to string so appended string
   })
   app.get("/j1",(req,res)=>{
      res.json(student.age)
   })
app.listen(6001)
```

Example : Write Express JS script to request server to display json object (Array of Objects) in table form on browser.

```
const expr=require("express")
const app=expr();
student={
   u1:[{name:"LJU",id:2},
   {name:"LJU1",id:3},
   {name:"LJU2",id:4},
   {name:"LJU3",id:5},
   {name:"LJU4",id:6}]}
      app.get("/student",(req,res)=>{
         res.set("content-type","text/html")
          res.write("<center><table cellspacing='5px' cellpadding='8px'
border='1px solid'><tr><th>Name</th><th>ID</th></tr>")
          for(i of student.u1){
             res.write("<tr><td>" + i.name + "</td>")
             res.write("<td>" + JSON.stringify(i.id) + "</td></tr>")
          }
         res.write("</table></center>")
         res.send()
      })
app.listen(6007)
```

Output:

Name	ID
LJU	2
LJU1	3
LJU2	4
LJU3	5
LJU4	6

Example : Write an express js script to define one JSON array of 3 objects having properties name and age. Short these objects according to age. If user request sorted names in url then all names along with age should be printed according to descending order of age. Also, display these sorted values on “Sort page” and display JSON object on “Home page”.

```
const expr = require("express")
const app = expr();
student = [{name:"abc",age:28},
        {name:"def",age:40},
        {name:"xyz",age:10}]
 //home page
app.get ("/",(req,res)=>{
   res.set ("content-type","text/html")
   res.send (student)
})
//sort page
app.get('/sort', (req, res) => { res.set ("content-type","text/html")
const sorted=student.sort((a, b) => b.age - a.age);
   for (i of sorted){
      res.write ("<center><h2>"+i.name+ " = " + i.age +"</h2></center>")
   }
res.send();
})
app.listen(3006)
```

/ Alternate approach of "sort page" More Specific Based on Demand which will not sort json object on home page after refreshing//

```
//sort page

app.get('/sort', (req, res) => {
    const sorted = student.slice().sort((a, b) => b.age - a.age);
   // or
   //  const sorted = [...student].sort((a, b) => b.age - a.age);
    let output = "";
    for (const k of sorted) {
        output += `<center><h2>${k.name} = ${k.age}</h2></center>`;
    }
    res.send(output);
});
```

Output:

On home page

[{name:"abc",age:28}, {name:"def",age:40}, {name:"xyz",age:10}]

On sort page (*Specific)

Def = 40
abc = 28
xyz = 10

Example : Set 404 status code on page not found error. (Ref*)

```
const expr=require("express");
const app=expr();
app.get("/",(req,res)=>
{
res.write("Home Page"); res.send();
})
app.get("/student",(req,res)=>
{
res.write("Student Page"); res.send();
})
app.use((req,res)=>
{
res.status(404).send("page not found");
})

// Insert app.use after all possible  path routing for page not found
app.listen(8081,()=>{console.log("server started");})
```

app.use() Method

What is app.use()

- In Express.js, app.use() is used to attach middleware functions to your application.
- A middleware is a function that runs during the request and response cycle.
- Flow of execution:
- Request → Middleware → Route → Response

When do we use app.use()

Use app.use() when you want something to run:

- Before every request
- Before specific routes
- For processing incoming data
- For adding features such as security, logging, or validation

Syntax :app.use(path, middlewareFunction)

- If no path is provided, it runs for all routes
- If a path is provided, it runs only for matching routes

Types of Middleware used in app.use()

JSON Middleware

app.use(express.json())
Used to parse JSON data from incoming requests and store it in req.body.

URL Encoded Middleware

app.use(express.urlencoded({ extended: true }))
Used to parse form data sent from HTML forms.

Cookie Parser Middleware

app.use(cookieParser())
Used to read cookies sent by the browser and access them using req.cookies.

Express Session Middleware

app.use(session({...}))
Used to store and manage user session data on the server using req.session.

Multer Middleware

app.use(upload.single("file"))
Used to handle file uploads from forms (like images, documents).

CORS Middleware

app.use(cors())
Used to allow communication between frontend and backend running on different domains or ports.

Static Files Middleware

```
app.use(express.static('public'))
```
Used to serve static files like HTML, CSS, JavaScript, and images from a folder.

Custom Middleware

```
app.use((req, res, next) => {
console.log(req.method, req.url);
next();
})
```
Used to add custom logic such as logging, authentication, or validation.

app.get('/') vs app.use('/') in Express.js

app.get('/'): Handles only GET requests to the exact / path.
Example: Only runs for http://localhost:3000/.

app.use('/'): Handles all HTTP methods for all paths starting with /.
Example: Runs for /, /contact, /about, /anything.

app.use() handles all HTTP methods, while app.get() handles only GET requests, and middleware parameters must follow (req, res, next) order.

Route parameters

Route parameters are named URL segments that are used to capture the values specified at their position in the URL. The captured values are populated in the **req.params** object, with the name of the route parameter specified in the path as their respective keys.

Route path: /calendar/:day/event/:ename
Request URL: http://localhost:6001/calendar/monday/event/birthday
req.params will return object: {"day":"monday","ename":"birthday"}

To use the dynamic routes, we should provide different types of routes. Using dynamic routes allows us to pass parameters and process based on them.

Here is an example of a dynamic route:

```
var express = require('express');
var app = express();
app.get('/:id', (req, res)=>{
 res.send('The id you specified is ' + req.params.id);
});
app.listen(3000);
```

To test this go to http://localhost:3000/**123**.

You can replace '123' in the URL with anything and the change will reflect in the response.

Note : To define routes with route parameters, simply specify the route parameters in the path of the route as shown below.

Example:

```
var expr = require("express");
var app = expr();
app.get('/user/:userId/test/:test', (req, res) => {
   req.params;
   res.send(req.params);
  });
  app.listen(5000)
```

Output: { "userId": "342", "test": "889" }

Request URL: http://localhost:5000/user/342/test/889

Output due to req.params: { "userId": "342", "test": "889" }

Task-1

Route path: '/things/:name/:id'

Request URL : http://localhost:3000/things/practice/12345.

Output:{“name”: “practice” , “id”:12345}

Task-2

Route path: /flights/:from-:to

Request URL: http://localhost:3000/flights/ADI-NYC

Output: { "from": "ADI", "to": "NYC" }

Query Parameters/req.query/req.body

Express.js is a web application framework for Node.js that is used to build web servers and APIs. When a client sends data to the server, Express provides different ways to access that data. The most common methods are through query parameters and the request body. These are accessed using req.query and req.body respectively.

Query parameters are key value pairs that are sent in the URL after a question mark. They are mainly used to send small amounts of data such as filters, search terms, or sorting options.

Characteristics of query parameters

- Query parameters are always received as strings
- They are visible in the URL
- They are mostly used with GET requests
- They are suitable for small amounts of data

Example of a URL with query parameters

http://localhost:3000/products?category=mobile&price=20000

In this example, category and price are query parameters.

In Express.js, query parameters are accessed using **req.query**.

❖ req.query

It is an object in Express.js that contains all query parameters sent in the URL.

Example
If the request URL is : /search?name=rahul&age=25
Then req.query will be :{ name: "rahul", age: "25" }

This object allows developers to easily access and use query data in applications.

```
app.get('/search', (req, res) => {
const name = req.query.name;
const age = req.query.age;
res.send("Name: " + name + ", Age: " + age);
});
```

❖ req.body

The request body is used to send data from the client to the server in HTTP requests such as POST, PUT. This data is not visible in the URL and can include large or structured data such as JSON.

In Express.js, this data is accessed using **req.body**.

Characteristics of req.body

- Data is not visible in the URL
- It can handle large and complex data
- Used mainly with POST, PUT methods
- Supports JSON and form data

Example of JSON data sent by client: {"name": "Rahul","age": 25}

Example

```
app.use(express.urlencoded({ extended: true }));
app.post('/users', (req, res) => {
const name = req.body.name;
const age = req.body.age;
res.send("Name: " + name + ", Age: " + age);
});
```

Middleware requirement
To use req.body, middleware must be enabled. Express provides built in middleware such as express.json() and express.urlencoded() to parse incoming data. Without this middleware, req.body will be undefined.

Example

```
app.use(express.json());
app.use(express.urlencoded({ extended: true }));
```

extended: false

- Only supports **simple (flat) data**
- Cannot handle nested objects
- Example → { "user": "test" }
- Uses Node.js built-in **querystring** module

extended: true

- Supports **nested objects & arrays**
- Cleaner and more useful structure
- Uses external **qs** library
- Example → { user: { name: "test" } }

Feature	express.json()	express.urlencoded()
Data type	JSON data	Form data (URL-encoded)
Example input	{ "name": "test" }	name=test&age=25
Parses into	req.body object	req.body object
Use case	APIs / backend services	Traditional web forms

Example-1

Create a web server that:

- **Accepts a GET request with a dynamic user ID in the URL provided "http://localhost:3000/user/44?name=test&age=23"**
- **Also accepts additional data (name, age) using query parameters**
- **Extracts:**
  - **id from route parameters**
  - **name and age from query string**
- **Returns a JSON response containing all received data**

```
const express = require("express");
const app = express();

// Route with params + query
app.get("/user/:id", (req, res) => {
   const userId = req.params.id;     // from URL path
   const name = req.query.name;      // from query string
   const age = req.query.age;
   res.json({
      message: "Data received",
      params: { id: userId },
      query: { name, age }   //JS  shorthand {name} returns={name: req.query.name }
   });
});
app.listen(3005, () => {  console.log("Server running on port 3005");});
```

Output :

```
{"message":"Data received","params":{"id":"44"},"query":{"name":"test","age":"23"}}
```

Example-2

Create a basic web application that:

- **Displays a form on the homepage to enter a user's name**
- **Sends the form data to the server using a get request**
- **The server should read the submitted name from the request body**
- **Finally, display a message confirming:**

```
const express = require("express");
const app = express();

// Homepage with form
app.get("/", (req, res) => {
   res.send(`<form action="/user" method="get">
         <input type="text" name="name" placeholder="Enter name" />
         <button type="submit">Create</button>
      </form>`);
});
// GET request
app.get("/user", (req, res) => {
   const name = req.query.name;
   res.send("User created with name " + name);
});
// Start server
app.listen(5504, () => {console.log("Server started on port 5504");});
```

Example -3

Create a basic web application that:

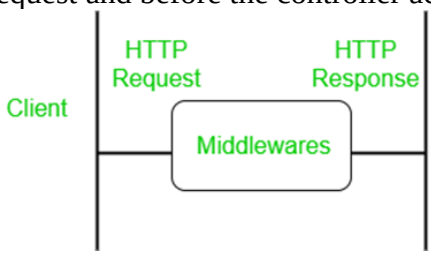
- **Displays a form on the homepage to enter a user's name**
- **Sends the form data to the server using a POST request**
- **The server should read the submitted name from the request body**
- **Finally, display a message confirming:**

```
const express = require("express");
const app = express();
// Middleware
app.use(express.urlencoded({ extended: true }));
// Homepage with form
app.get("/", (req, res) => {
   res.send(`<form action="/user" method="post">
         <input type="text" name="name" placeholder="Enter name" />
         <button type="submit">Create</button>
      </form>`);
});
// POST request
app.post("/user", (req, res) => {
   const name = req.body.name;
   res.send("User created with name " + name);
});
// Start server
app.listen(5504, () => { console.log("Server started on port 5504");});
```

Middleware

What is Middleware in Express JS

Express.js is a routing and Middleware framework for handling the different routing of the webpage and it works between the request and response cycle. Middleware gets executed after the server receives the request and before the controller actions send the response.



Middleware is a request handler that allows you to intercept and manipulate requests and responses before they reach route handlers. They are the functions that are invoked by the Express.js routing layer.

It is a flexible tool that helps in adding functionalities like logging, authentication, error handling, and more to Express applications.

Middleware Syntax
The basic syntax for the middleware functions is:

Middleware Syntax

```
const middlewareName = (req, res, next) => {
   // process request
   next();
};
```

Using Middleware

```
app.use(middlewareName);
```

OR

```
app.get("/route", middlewareName, (req, res) => {
   res.send("Response");
});
```

- req → request data
- res → response
- next() → move to next middleware

Middleware is a function with (req, res, next) that processes request before response.

- Middleware functions take 3 arguments: the request object, the response object, and the next function in the application's request-response cycle, i.e., two objects and one function.
- Middleware functions execute some code that can have side effects on the app and usually add information to the request or response objects. They are also capable of ending the cycle by sending a response when some condition is satisfied. If they don't send the response when they are done, they start the execution of the next function in the stack. This triggers calling the 3rd argument, next().

- The middle part (req,res,next)=>{} is the middleware function. Here we generally perform the actions required before the user is allowed to view the webpage or call the data and many other functions. So let us create our own middleware and see its uses.

Advantages of using middleware:

- Middleware can process request objects multiple times before the server works for that request.
- Middleware can be used to add logging and authentication functionality.
- Middleware improves client-side rendering performance.
- Middleware is used for setting some specific HTTP headers.
- Middleware helps for Optimization and better performance.

Types of Middleware

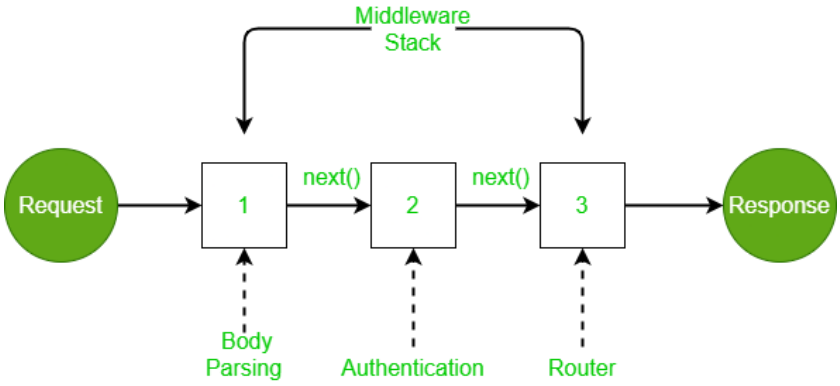
Express JS offers different types of middleware and you should choose the middleware on the basis of functionality required.

Type	Syntax	Application	Why it is used	When to use
Built – in middleware	express.json()	JSON	Read JSON body	APIs
	express.urlencoded()	URL Encoded	Read form data	Forms
	express.static()	Static Files	Serve files	Websites
Third party middleware	cookie-parser()	Cookies	Read cookies	Auth, Sessions
	express-session()	Sessions	Store user session	Login systems
	multer()	File upload	Handle file uploads	Forms with files
	cors()	CORS	Allow cross-origin	Allow Frontend to call Backend
Custom middleware	(req, res, next) function	Custom	Custom logic	Logging, Authentication

Middleware Chaining: Middleware can be chained from one to another, Hence creating a chain of functions that are executed in order. The last function sends the response back to the browser. So, before sending the response back to the browser the different middleware process the request.

The **next()** function in the express is responsible for calling the next middleware function if there is one.

Modified requests will be available to each middleware via the next function –



Middleware chaining example

In the above case, the incoming request is modified and various operations are performed using several middleware, and middleware is chained using the next function. The router sends the response back to the browser.

Using app.use()

app.use(path, (req, res, next) => {}, (req, res, next) => {}..)

app.get(path, (req, res) => {})

or

use as Route Middleware

app.get/post(path, (req, res, next) => {}, (req, res, next) => {}, …. (req, res) => {})

Note:

Use app.use() → when logic is **common**
Use route middleware → when logic is **specific**

Example 1

```
var expr = require("express");
var app = expr();

//Method 1
const cb=(req,res,next)=>{
   console.log("Initalized");
   res.set("content-type","text/html")
   res.write("<strong>First</strong>");
   next();
}
const cb1=(req,res,next)=>{
   res.write("<p>Addition = " + (5+5) + "</p>");
   next();
}
app.use("/ee",cb,cb1);
```

```
app.get("/ee",(req,res)=>{
   res.write("<h1>Hello Welcome to LJU</h1>");
   res.send();
});

//Method 2
app.get( "/",  (req, res, next) => {
     console.log("request received on"+new Date());
     next();
   },
   (req, res, next) => {
      res.set("content-type","text/html")
      res.write("Hello");
      next();
     },
   (req, res) => {
     res.write(`<div>
     <h2>Welcome to LJU</h2>
     <h5>Tutorial on Middleware</h5>
   </div>`);
   res.send();
   });
app.listen(6001)
```

Output: Observe Sequence by running both the methods

Example 2

Create an application using Express.js where:

- **Multiple middleware functions add student name, college, and marks**
- **Data is passed using req**
- **Final route displays all values**

```
const express = require("express");
const app = express();

// Middleware 1: Add student name
const addName = (req, res, next) => {
   req.name = "Samarth";
   console.log("Name added");
   next();
};

// Middleware 2: Add college name
const addCollege = (req, res, next) => {
   req.college = "LJ University";
   console.log("College added");
   next();
};
```

```
// Middleware 3: Perform calculation
const addMarks = (req, res, next) => {
   req.total = 50 + 40;
   console.log("Marks calculated");
   next();
};

// Route with middleware chain
app.get("/student", addName, addCollege, addMarks, (req, res) => {
   res.send("Student Name: " + req.name +
      "<br>College: " + req.college +
      "<br>Total Marks: " + req.total );
});

// Start server
app.listen(6001, () => {console.log("Server running on port 6001");});
```

Type in browser: http://localhost:6001/student

Output:

Browser:
Student Name: Samarth
College: LJ University
Total Marks: 90
Console:
Server running on port 6001
Name added
College added
Marks calculated

Example 3

Develop a web application that simulates a college classroom entry system using middleware:

- **When a student accesses the /class route:**
  1. **Log that the student entered the campus**
  2. **Verify whether the student has a valid ID card**
  3. **If valid → allow entry and display a welcome message**
  4. **If not valid → deny access**

```
const express = require("express");
const app = express();

// Middleware 1: Entry Log (like college gate entry)
const entryLog = (req, res, next) => {
   console.log("Student entered campus");
   next();
};
```

```
// Middleware 2: ID Check
const checkID = (req, res, next) => {
    const hasID = true; // change to false to test

    if (hasID) {
        req.student = "A"; // add student name
        console.log("ID verified");
        next();
    } else {
        res.send("Access Denied: No ID Card");
    }
};

//  Apply middleware chain
app.use("/class", entryLog, checkID);

//  Final Route (Teacher)
app.get("/class/data", (req, res) => {
    res.send(`Welcome ${req.student} to the class`);
});

// Start server
app.listen(3000, () => {console.log("Server running on port 3000");});
```

In browser: http://localhost:3000/class/data

Output:
Browser:
Welcome A to the class
Console:
Server running on port 3000
Student entered campus
ID verified

Explanation:

- Middleware function must have **(req, res, next)**
- Must call **next()** to move to next middleware
- If next() is not called → request will stop
- If response is sent using **res.send()**
- Then do **NOT call next() after that**
- Middleware must be **defined before route**
- app.use("/class", entryLog, checkID); // first
- app.get("/class/data", ...); // then